

2ND ROUND SET-B

Using AI to answer the questions will result in disqualification.

Question number -1

Outline an algorithm for finding a minimum spanning tree (MST) in a distributed system.

Follow-up: In what real-world ML scenario would you use MSTs (e.g., sensor network feature selection)?

Question number -2

Given a large stream of user interactions, design a data structure for $O(\log n)$ range-sum queries and updates, supporting both exact and approximate aggregation for downstream feature generators.

Follow-up: How would you parallelize updates for distributed feature stores?

Question number -3

You are tasked with designing an ML model cache. Explain how a splay tree helps optimize access patterns and what the trade-offs are compared to LRU caches.

Follow-up: When would splaying harm latency in serving predictions?

Question number -4

Build a system to support efficient merging and querying of connected user components in a social graph as edges are added or deleted for clustering in ML.

Follow-up: How does union by rank and path compression speed up large-scale ML graph operations?